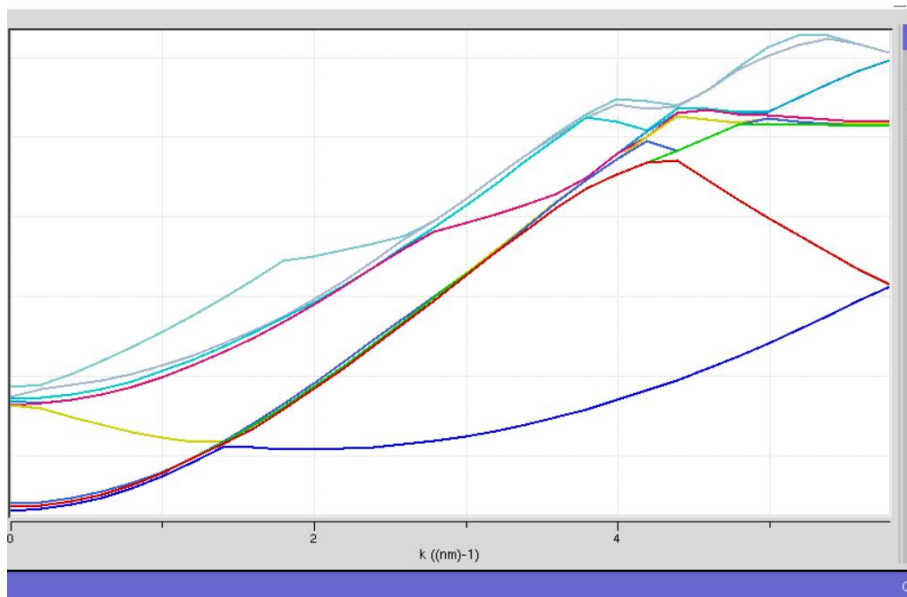
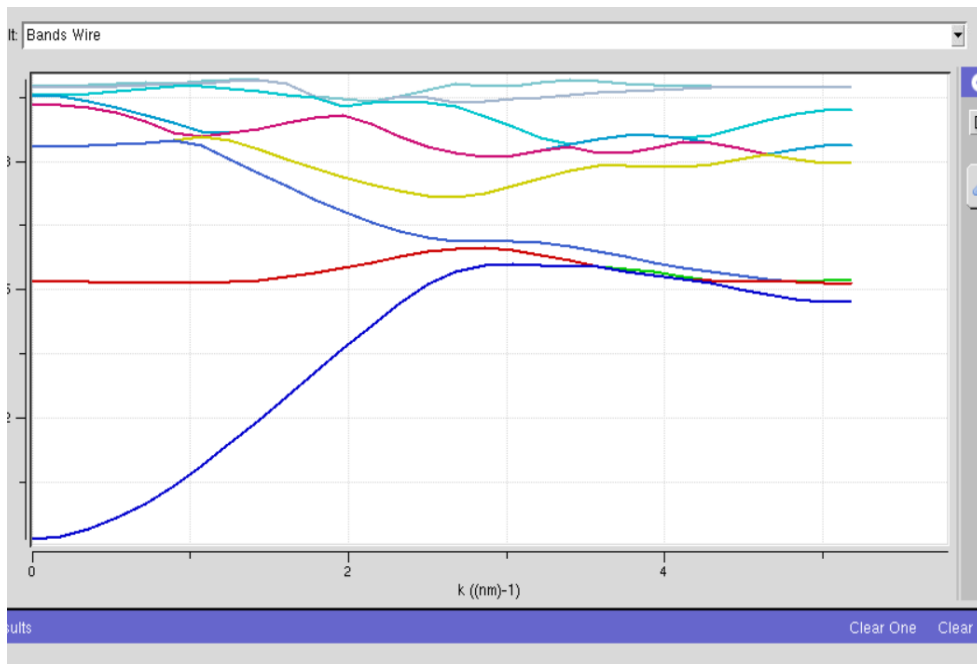


Experiment 5: Band Structure Simulation for Graphene and CNTs using NanoHub

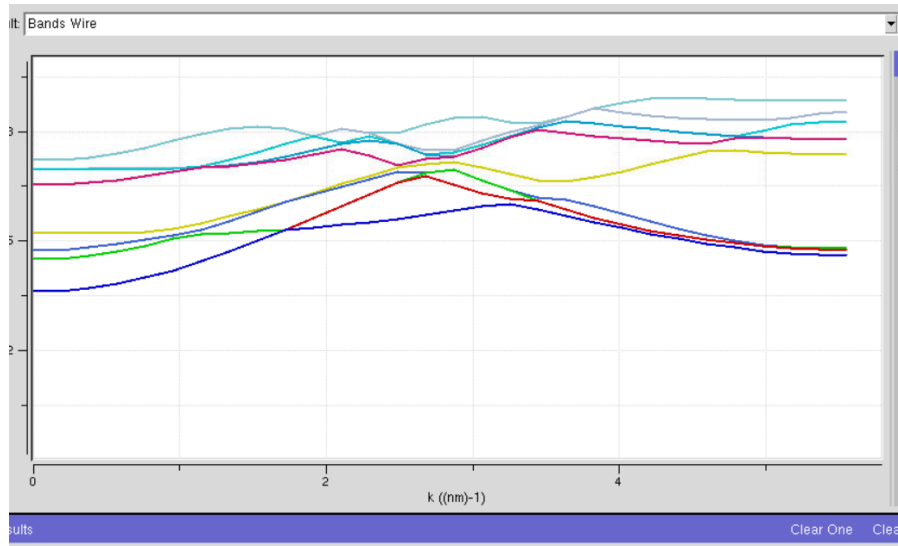
Output:



Test case:01 Simulate the band-structure of InAs circular nanowire



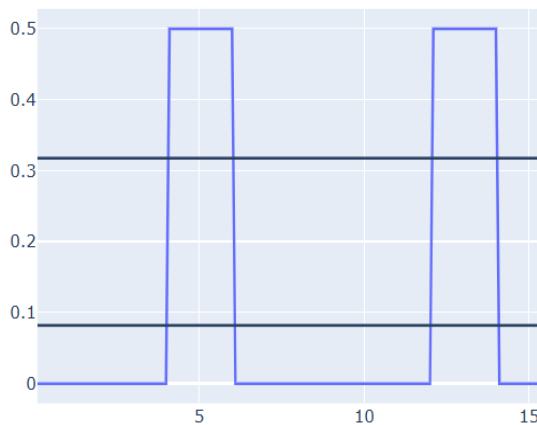
Test Case: 02 Simulate the band-structure of AlGaAs circular nanowire



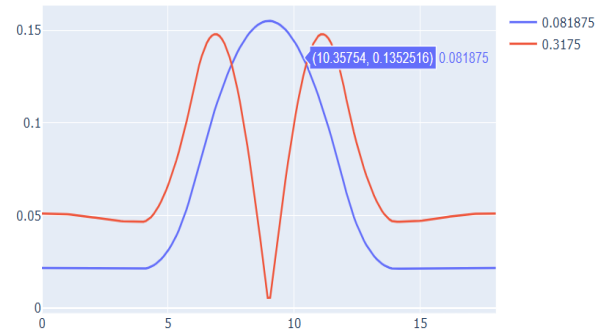
Experiment 6: Demonstration of Quantum Tunneling

Output:

Potential Energy vs Distance



Wave-functions from tight-binding



The graph shows a periodic potential energy profile, where the high rectangular regions represent potential barriers and the low regions represent potential wells. This periodic arrangement models the repeating atomic lattice structure in a crystal.

Test Case :01 Analysis the transmission co-efficient Vs Energy characteristics on refined grid



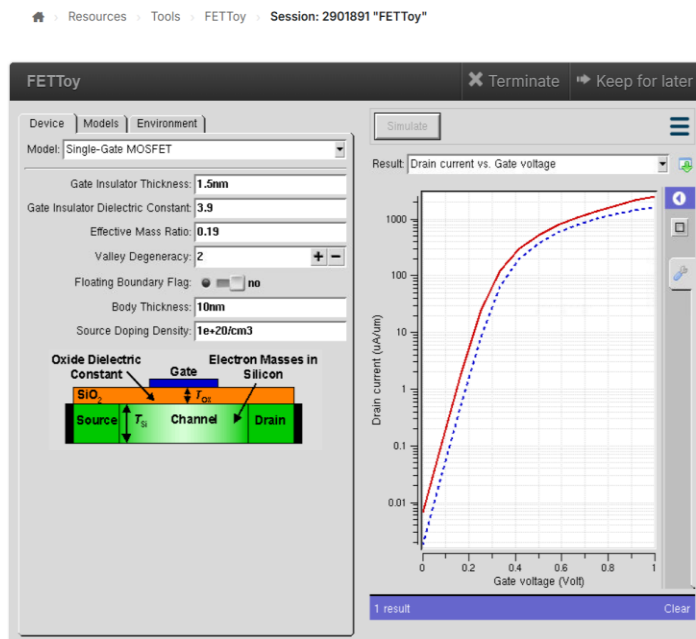
The graph shows that the transmission coefficient increases with energy, indicating that electrons are more likely to tunnel through the periodic potential barriers at higher energies. The close overlap of the Transfer Matrix and Tight-Binding method curves confirms that both methods produce consistent and accurate results.

Test Case:02 Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid



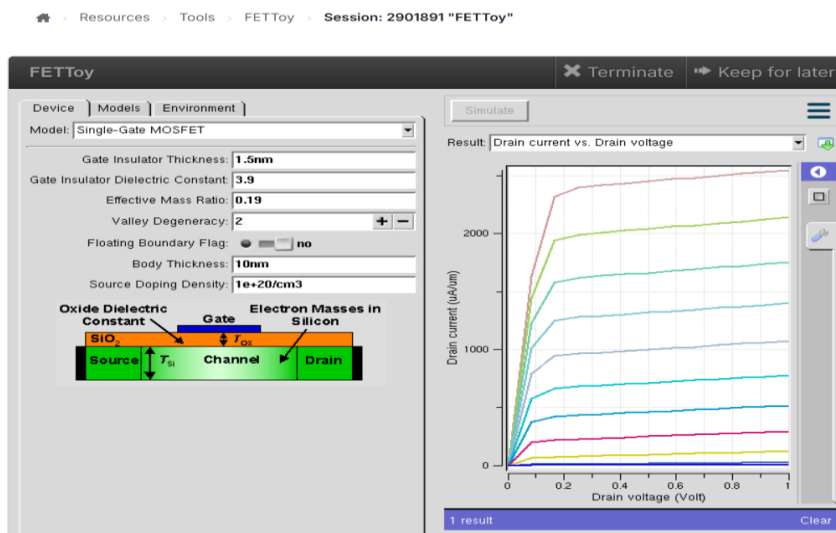
Both graphs show that the reflection coefficient decreases as the electron energy increases, indicating that electrons are less likely to be reflected and more likely to transmit through the potential barriers at higher energies. The close overlap of the Transfer Matrix and Tight-Binding method curves in both the refined and homogeneous grids demonstrates the accuracy, consistency, and reliability of the two methods for analyzing electron transport in periodic structures.

Experiment 7: MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub



The graph shows that the drain current increases rapidly as the gate voltage increases, indicating enhanced channel formation and improved electron conduction. The close agreement between the simulated (solid red) and reference (dashed blue) curves validates the accuracy of the MOSFET model and confirms its expected electrical behavior.

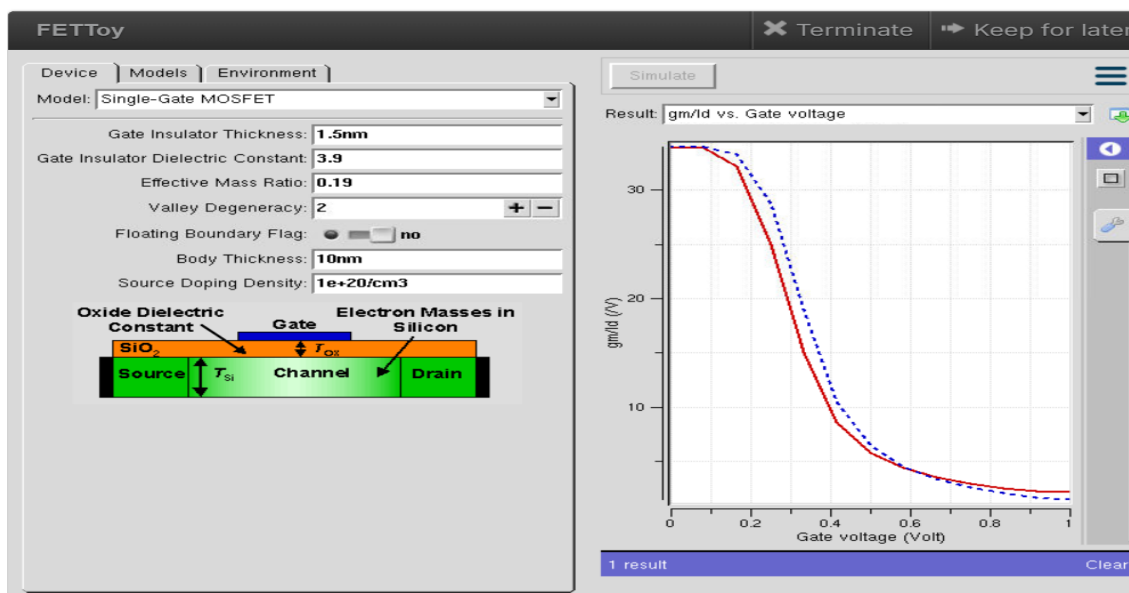
Test Case :01 Increase in V_G : • Observation: As V_G increases, the drain current (I_D) also increases exponentially due to the MOSFET entering the saturation region earlier.



As the drain voltage (V_D) increases, the drain current (I_D) rises rapidly in the linear region and then becomes nearly constant after reaching saturation. Higher gate voltages produce higher saturation currents, demonstrating stronger channel conduction and improved current-driving capability.

Test case: 02 Variation in V_D :

- Observation: Drain current (I_D) saturates for higher V_D at a fixed V_G



As the gate voltage (V_G) increases, the g_m/I_D ratio decreases, indicating reduced transconductance efficiency at higher operating voltages. The close agreement between the simulated (solid red) and reference (dashed blue) curves validates the MOSFET model and accurately represents its switching behavior.